23 January 2026

Free Energy Simulations on a Biomimetic Glucose Receptor: Understanding the Selectivity of GluHUT

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Abstract

The bicyclic glucose receptor GluHUT represents a significant advance in supramolecular chemistry, binding glucose in water with an association constant of approximately 18,000 M^{-1}, and showing exceptional selectivity. Elucidating the molecular basis of this binding selectivity could be valuable for the rational design of receptors for other carbohydrate targets. The intricate dynamics of the binding process are probed here through extensive molecular dynamics (MD) simulations and enhanced sampling techniques to analyse the complexation of glucose, galactose, and fructose with GluHUT in water. Through performing umbrella sampling simulations, we obtained an atomic-level view of the entire binding event, revealing both the thermodynamic drivers of selectivity and the transition states involved. In-depth analysis of hydrogen bonding patterns, noncovalent interactions, and solvent reorganization showed that even minor structural variations in sugars can produce significant differences in binding affinity. These findings, supporting the computed binding free energies that closely mirror experimental data, provide a mechanistic rationale for GluHUT's remarkable specificity toward glucose.

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ARTICLE

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Received 00th January 20xx,
Accepted 00th January 20xx

DOI: 10.1039/x0xx00000x

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The bicyclic glucose receptor GluHUT represents a significant advance in supramolecular chemistry, binding glucose in water with an association constant of approximately 18,000 M^{-1}, and showing exceptional selectivity. Elucidating the molecular basis of this binding selectivity could be valuable for the rational design of receptors for other carbohydrate targets. The intricate dynamics of the binding process are probed here through extensive molecular dynamics (MD) simulations and enhanced sampling techniques to analyse the complexation of glucose, galactose, and fructose with GluHUT in water. Through performing umbrella sampling simulations, we obtained an atomic-level view of the entire binding event, revealing both the thermodynamic drivers of selectivity and the transition states involved. In-depth analysis of hydrogen bonding patterns, noncovalent interactions, and solvent reorganization showed that even minor structural variations in sugars can produce significant differences in binding affinity. These findings, supporting the computed binding free energies that closely mirror experimental data, provide a mechanistic rationale for GluHUT's remarkable specificity toward glucose.

Introduction

The selective binding of carbohydrates in aqueous solution presents a significant challenge. Saccharides are highly hydrophilic, interacting strongly with water, and share generally similar structures. A useful receptor must differentiate between subtly different targets, whilst also overcoming competition from solvent. Glucose is a ligand of particular biological importance, due to its central role in metabolism and, medically, its relevance to diabetes. Effective glucose receptors can lead to sensing systems which can help to advance the next generation of diabetes therapies.[1,2,3] With nature as inspiration, our group in Bristol has synthesised a variety of carbohydrate receptors with a focus on glucose and other "all-equatorial" carbohydrates. The central design architecture has been christened a "temple", with pillars rich in hydrogen bonding functionalities such as ureas or amides, combined with hydrophobic 'roof' and 'floor' units to accommodate the non-polar regions of the carbohydrates.[4,5,6,7,8,9,10,11] The most recent, and successful, iteration of this architecture is GluHUT (Glucose binding HexaUrea Temple, Figure 1a).[12] With binding constants to glucose of K_a = 17,300 – 18,300 M^{-1} in phosphate buffered saline solution, GluHUT was a major breakthrough, surpassing previous systems by two orders of magnitude. It is also highly selective, favoring glucose over galactose by a factor of 100 and showing negligible affinities for non-carbohydrate targets. This meant that the molecule could be considered for use in medicine, suitable for both sensing and therapeutic applications. Its potential has recently been demonstrated in a glucose-responsive insulin which protects against hypoglycemia in animal studies.[13] Glucose sensitivity was achieved by attaching a GluHUT receptor to an insulin peptide at one end, and a β-glucoside to the other. Intramolecular binding dominates at low glucose concentrations, depressing insulin activity, whilst at higher concentrations the intramolecular interaction is overcome by glucose binding and insulin activity is restored.

Computational modelling was important in the design of GluHUT, but was limited to Monte Carlo molecular mechanics (MM) implemented through a standard commercial package.[15] The calculations predicted an "all -NH in" conformation for the receptor core (Figure 1a-b), and this was supported by 2D NMR experiments.[12] It is understood that this well-defined structure is key to the binding properties of GluHUT, but detailed further investigation is needed to study the more intricate features that create such an ideal microenvironment for glucose, and also why other sugars bind much less strongly (see Figure 1c for experimental binding affinities). Understanding why, for instance, galactose binds 100x less tightly than glucose, will help in designing the next generation of carbohydrate receptors targeting other biologically important ligands.

Host-guest binding in other systems has been studied with various computational methods, including quantum mechanical calculations and MM molecular dynamics (MD)

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Supplementary Information available

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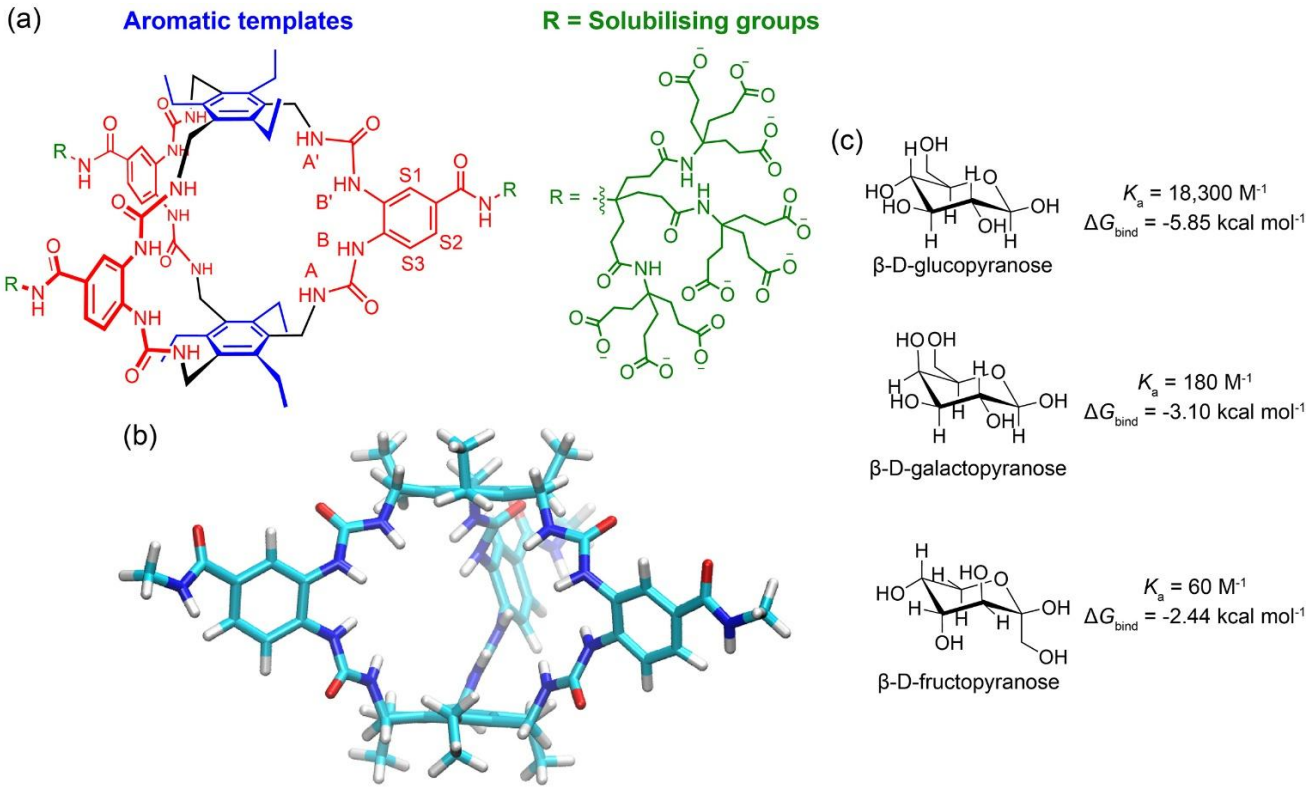

Figure 1. (a) Chemical structure of GluHUT. (b) Structure of the truncated GluHUT receptor model used in this work. (c) Carbohydrate substrate molecules studied in this work. The binding constant K_a values in water were obtained from Tromans et al.,[12] and converted to binding free energy at 300 K. All three carbohydrates exist as mixtures of tautomers. For glucose and galactose, the major forms are the β-pyranose as shown. For fructose, both pyranose and furanose forms are significant; the major anomer is shown here for each.[14]

simulations.[16,17,18,19,20] An advantage of MM MD simulations is that enhanced sampling techniques can be utilised on explicitly solvated systems, allowing for the exploration of larger systems, such as macromolecular complexes in water. In this work, we applied umbrella sampling, a path-based enhanced sampling method where harmonic restraints are applied at closely overlapping windows across a predefined reaction coordinate, yielding a potential of mean force (PMF), giving complexation affinities.[21,22,23] We found that the relative computed binding free energies were in good agreement with the experimental values for the monosaccharides studied in this work. Analysis of the trajectories shows that hydrogen bonding (H-bonding) and solvent effects explain the origins of the binding selectivity. The results suggest that these methods can be used to inform the design of new receptors aimed at other targets of biological importance.

Results and discussion

Selectivity of GluHUT for Binding Glucose Reproduced with Free Energy Simulations. To study GluHUT's binding properties, four different guest molecules were investigated in this work: β-D-glucose, β-D-galactose, β-D-fructopyranose and β-D-fructofuranose (further referred to as glucose, galactose, fructopyranose and fructofuranose). For glucose and galactose, the tautomers chosen for study were the β-pyranose forms, which predominate in aqueous solution (Figure 1c). In the case of glucose there is evidence that the β-pyranose is the major substrate for GluHUT, and it seems likely that the same is true for galactose.[12] In the case of fructose, the form which is bound is less certain as both furanose and pyranose tautomers are present in water. We therefore chose to study both, as the major (β) anomers (Figure 1c).[14]

Prior to umbrella sampling, a series of steered MD (SMD) simulations of the unbinding process of each sugar from the receptor were carried out. Starting with a bound conformation, these simulations involved 'pulling' the monosaccharide out of the GluHUT host at a constant speed (see Methods for details on the reaction coordinate). In regions of strong interaction, more force is required to displace the guest molecule, and these forces can be plotted as a work profile (Figure S1). Although this highlights regions of strong interaction, and shows clear energy barriers, the primary use of these trajectories was to generate snapshots for subsequent umbrella sampling simulations.

The PMFs obtained from these simulations show good agreement with the trends in the experimentally derived binding energy values (Figure 2). The computed binding free energy for glucose was –12.5 kcal mol^{-1}, experimentally measured at –5.9 kcal mol^{-1}, and for galactose –8.1 kcal mol^{-1}, experimentally measured at –3.1 kcal mol^{-1}. Both sugars have

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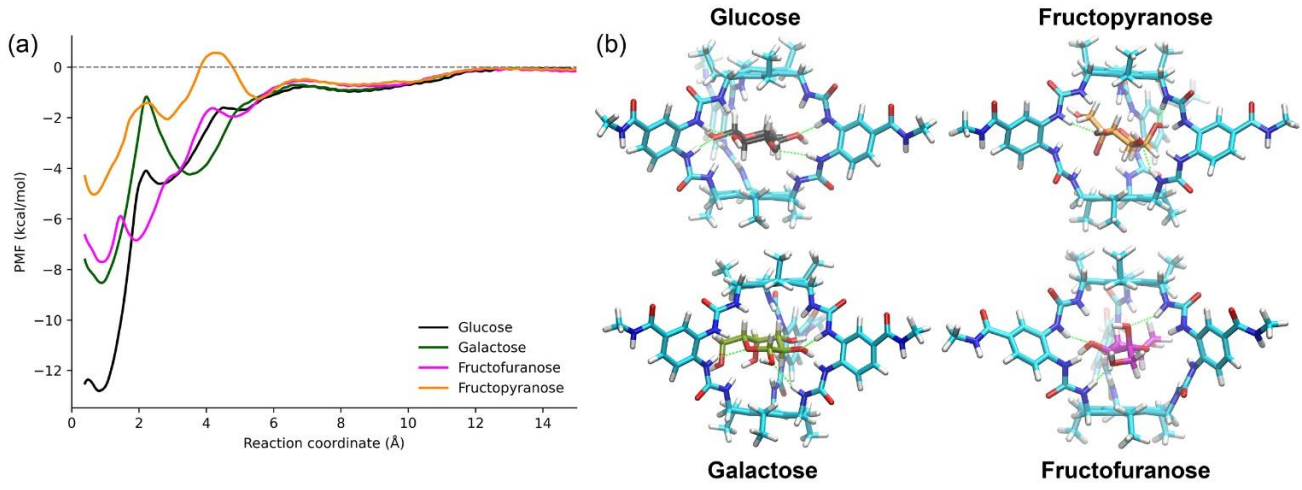

Figure 2. (a) The average potential of mean force (PMF) for glucose, galactose, fructopyranose and fructofuranose binding to GluHUT obtained from three independent umbrella sampling simulations at 300 K. See Figure S2 for independent PMFs used to construct average PMFs. The binding free energies are calculated by subtracting the energy of the global minimum in the PMF from the energy in the bulk. The simulation concentration was 14 mM. See Methods for concentration corrected affinities. (b) Representative structures of bound states.

similar shaped PMFs, with a deep minimum before reaction coordinate (RC) value around 2 Å, but a significant energy barrier was found for galactose at RC ≈ 2 Å – 2.5 Å. The unbinding of galactose was further explored by performing additional umbrella sampling simulations shown in Figure S3.

For fructose, fructopyranose has significantly weaker interactions with the GluHUT cavity, with a computed binding free energy of –5.0 kcal mol^{-1}, and a PMF possessing a clear energy barrier at RC ≈ 4.5 Å. Fructopyranose is more mobile in the cavity than glucose or galactose, consistent with the smaller computed binding free energy. On the other hand umbrella sampling simulations for fructofuranose yielded a significantly more negative binding free energy of –7.7 kcal mol^{-1}. These results suggest that fructofuranose, present as 20% of the tautomeric mixture at equilibrium, might be the predominant binding partner with GluHUT in solution.

To understand the origin of these differences in affinity, the simulations were first analysed using the linear interaction energy (LIE) method.[24] This showed that electrostatic interactions are the dominant driver for binding of all four sugars, with glucose having the largest contribution of –46 kcal mol^{-1} in the bound state, and fructopyranose a significantly smaller contribution of approximately –30 kcal mol^{-1}, see Figure S4 for details. The bound states were further analysed through computing water occupancy in the cavity and H-bonding interactions for receptor-ligand, receptor-water and ligand-water.

GIST Uncovers Entropy Cost of Bound Waters in GluHUT. Before considering the effects of water on receptor-ligand complexes, we first examined the empty receptor. Operating in water, GluHUT's binding energetics are profoundly influenced by solvent reorganisation, where cavity-resident water molecules act both as competitors and thermodynamic modulators.[25,26] Number density calculations on unrestrained simulations of the empty receptor revealed that the cavity maintains an average hydration shell of seven water molecules, predominantly arranged in "horseshoe-shaped" distributions around the urea -NH groups (Figure 3a). These tightly bound waters formed persistent H-bonds with polar urea motifs, measured at an average of 4 per frame, while avoiding hydrophobic roof/floor regions. Displacement of these ordered waters during monosaccharide binding provides potential entropy gains, as shown by grid inhomogeneous solvation theory (GIST).[27]

The total entropy ($T\Delta S_{tot}$) contribution to the hydration free energy derived from GIST analysis quantifies the first-order entropic penalty experienced by water molecules in each voxel of the grid, encompassing both translational ($T\Delta S_{trans}$) and orientational ($T\Delta S_{ori}$) components. These values, given in Figure 3, measure how much the ordering of water inside GluHUT deviates from bulk behaviour, reflecting the loss of configurational freedom when water molecules interact with the host. A large negative entropic penalty signifies that water in that region is more ordered and constrained than in bulk water, possessing fewer accessible microstates. In contrast, positive or less negative terms indicate water with greater freedom, behaving more similarly to bulk solvent. Through this analysis, regions in the host cavity with highly negative total water entropy can be identified as sites where water molecules are likely to be favourably displaced upon guest binding. The analysis reveals six regions of low-entropy water within the cavity, paired with the six urea groups in GluHUT and reflecting H-bonding to the urea -NHs (Figure 3b, orange and magenta isosurfaces). A seventh region in the centre of the cavity appears at the less negative level (magenta isosurface). Displacement of these ordered waters by a single, relatively rigid carbohydrate molecule would substantially increase the system's entropy, making such replacement thermodynamically favourable from an entropic perspective.

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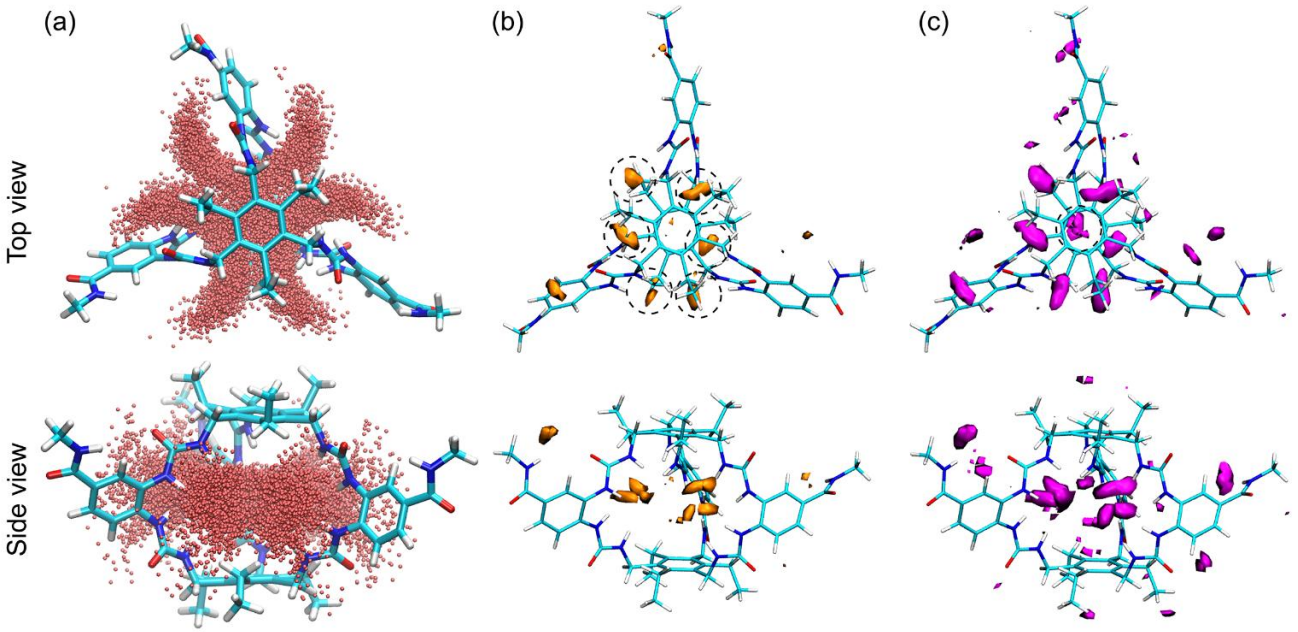

Figure 3. (a) Top and side views of the water occupancy in the empty receptor. (b) GIST isosurfaces showing the total entropic contribution to hydration free energy as a sum of translational and orientational entropic contributions derived from water positional and angular distributions. Isosurfaces represent water sites at two levels of negative (unfavourable) entropy, (b) the orange surfaces being more negative (–0.1 kcal/mol/water) than the (c) magenta surfaces (–0.06 kcal/mol/water). Both sets of surfaces are shown from two perspectives.

Glucose Completely Expels Water from GluHUT's Cavity. Radial distribution function (RDF) analysis of the complexes, and the empty receptor, showed that glucose achieved near-complete dehydration of the cavity, with galactose having slightly more water present. Both tautomers of fructose permitted some water occupancy (~2.5 molecules) in the cavity (Figure 4a). Whilst RDFs give an insight into the probability of finding water in the cavity, measuring the number of H-bonds can help to quantify this. The H-bond frequency between water oxygens and urea protons revealed an average of only 0.1 H-bonds between the cavity and water when glucose was bound, whilst this number was 0.9 for galactose and fructofuranose. Fructopyranose had the most competition with water, at 1.4 H-bonds formed, and this can be visualised by an increased density of water at the edge of the cavity interacting with two adjacent urea groups, competing for H-bonds with the sugar (Figure 4b).

Host-guest Hydrogen Bonding Dominates Selectivity. Based on the original modelling of GluHUT, H-bonding between host and guest is expected to be a key determinant of binding selectivity.[12] Given that our simulations are fully solvated, the analysis of hydrogen bonding of these systems need not be limited to host-guest interactions, but also guest-solvent and host-solvent. This in totality can help rationalise the phenomena we see here, not only highlighting the origins of selectivity for glucose, but the interactions which lead to

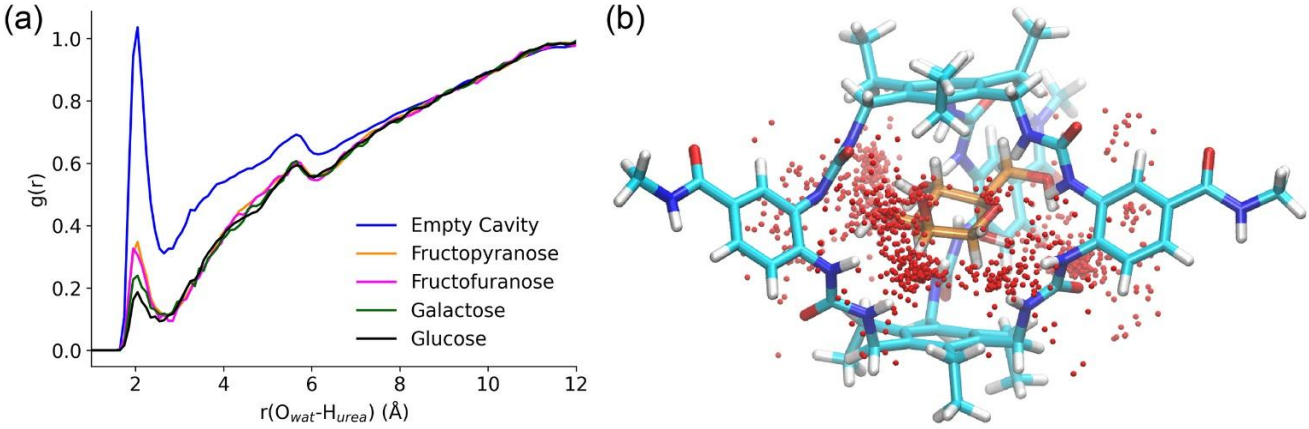

Figure 4. (a) The radial distribution function (RDF) of water oxygen atoms calculated from each of the urea protons (averaged) within GluHUT's cavity, for each of the four sugar complexes, and the empty cavity. (b) Positions of the nearest 3 water oxygens, per frame, to the centre of the cavity, for the unrestrained fructopyranose simulations. Note the lower front (left and right) and upper left urea groups are interacting with waters outside the cavity, a result of the poor interaction of fructopyranose with the cavity.

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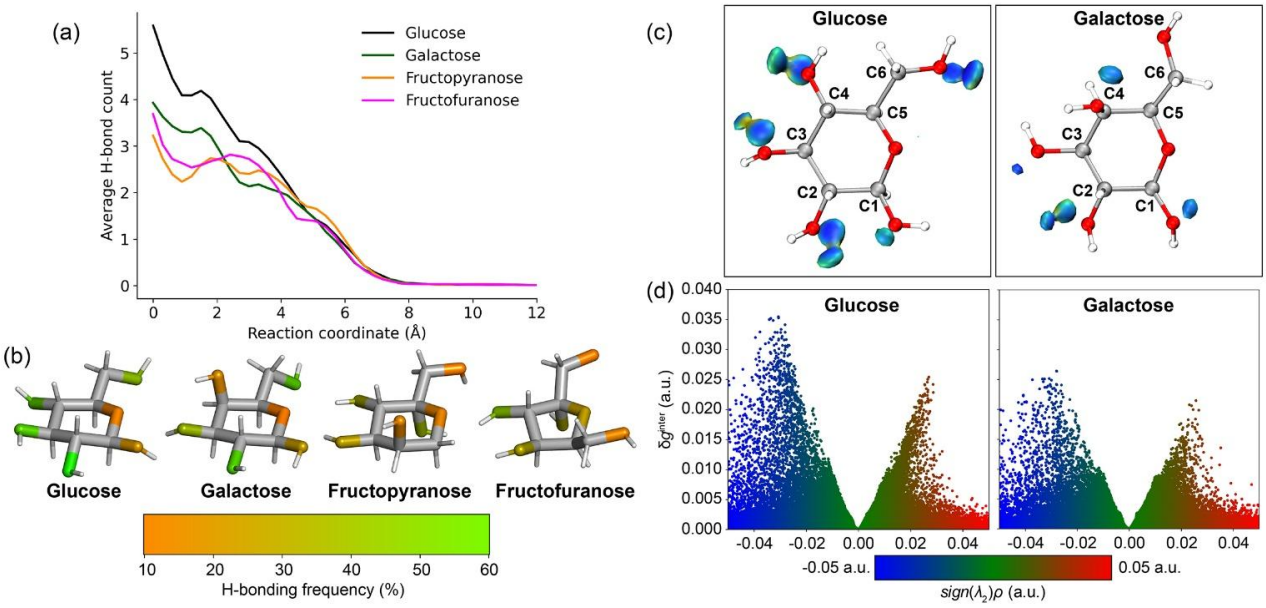

Figure 5. (a) H-bonding between GluHUT and each guest, along the binding reaction coordinate, averaged from 3 umbrella sampling runs. (b) Colour-coded H-bond frequency scores for each of the sugar oxygens (as H-bond acceptors) with GluHUT. (c) Sign(λ_2)ρ coloured isosurfaces of δg_{inter} = 0.02 a.u. from independent gradient model (IGM) analyses of glucose and galactose with GluHUT. The blue isosurfaces indicate strong attractive interactions, such as H-bonds, green isosurfaces indicate weaker noncovalent interactions such as van der Waals, whilst red indicates repulsive interactions, such as steric clashes. Only the carbohydrate guest is shown for clarity. (d) IGM scatter plots showing sign(λ_2)ρ against δg_{inter} descriptor for glucose and galactose in GluHUT.

transition states. To start, host-guest H-bonds were characterised across the entire PMF, using umbrella windows between RC = 0 Å and 15 Å. The results depicted in Figure 5a show that glucose has the strongest H-bonding with the receptor in the bound state, averaging 5.5 H-bonds between the sugar oxygens and the urea -NH atoms. Galactose had fewer H-bonds (around 4), followed by fructofuranose and finally fructopyranose. For more details of this H-bonding analysis, see Figure S5.

Clearly, the structural differences between the four sugars are the key to the differences in binding free energy. Our results indicate that H-bonding is the predominant driver for binding. Therefore, further (unrestrained) simulations of the bound complexes were performed (see Methods for details). The H-bonding between each sugar oxygen and the receptor was characterised, and scores were assigned to each oxygen, based on the frequency of H-bonds with GluHUT. Colour-coded representations of each sugar show which oxygens interacted most with the receptor, with orange and green indicating lower and higher percentages of interaction, respectively (Figure 5b). Fructopyranose, which has a more globular structure than the other sugars, showed relatively low H-bonding: no oxygens show strong H-bonding, suggesting a mismatch with the cavity. Fructofuranose showed relatively strong H-bonding to the oxygens directly attached to its furanose ring, while the CH_2-OH groups interacted less frequently. This may account for the tighter complex, given these groups are more flexible and less sterically hindered within the cavity, than for the bulky fructopyranose.

The most interesting comparison is between glucose and galactose, given their structures only vary by the stereochemistry of -OH in the C4 position, see Figure 5c. These two sugars interacted significantly more with the receptor, though with some clear differences between them. The axial -OH of galactose shows a relatively low H-bond frequency with the receptor, whilst the equatorial -OH of C4 for glucose frequently engaged in H-bonding. Noncovalent interaction (NCI) analysis revealed fine differences in the H-bonding of glucose and galactose to GluHUT (Figure 5c).[28,29] Namely, IGM isosurfaces confirmed strong H-bonding of the C4-OH of glucose with GluHUT. In contrast, for galactose this interaction of the axial C4-OH was much less prevalent. The IGM scatter plots depicted more negative (blue) values indicating a larger number of attractive noncovalent interactions (specifically H-bonds) for glucose compared to galactose. See Methods for details.

Insights from Sugar-water Interactions. Host-guest hydrogen bonds are clearly important in describing selectivity, but interactions with water must also be considered. Therefore, per-oxygen H-bond mapping approach was carried out for both glucose and galactose, where water acted as both the donor and the acceptor. It was found that the sugar oxygens were largely H-bond acceptors in the bound state, rarely donating to water. From unrestrained simulations, glucose formed around 4 hydrogen bonds with water compared to ~3 for galactose. Furthermore, the axial C4-OH of galactose formed almost no H-bonds with water, whilst the corresponding -OH for glucose was 0.7 on average (see Figure S6 for details). These enhanced interactions with solvent enable glucose to maintain stronger simultaneous H-bond networks with both receptor and solvent, thereby amplifying selectivity.

Besides absolute binding affinity, the other clear distinction in the PMFs of glucose and galactose was the pronounced barrier

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at RC ≈ 2 Å (Figure 2a and Figure S6). A combined host-guest and guest-water H-bond analysis was performed to investigate the origins of this barrier (see Figure S6 for details). It was found that in this region, glucose formed 2.6 H-bonds with the receptor, and 3.9 H-bonds with water. Galactose formed 1.8 H-bonds with the receptor, and 3.7 with water. Again, the axial -OH of galactose formed less H-bonds overall. Such analysis revealed fewer favorable interactions for galactose, likely attributing to the presence of a barrier.

Conclusions

Our simulations of GluHUT's sugar binding mechanisms reveal the molecular basis of glucose recognition and specificity. Molecular dynamics simulations reproduce the remarkable selectivity of this receptor, helping to provide a comprehensive understanding of how subtle structural differences dramatically influence binding interactions. They show four critical aspects of GluHUT's binding behavior. First, enhanced sampling binding free energy simulations showed good agreement with the trends in experimental binding affinities, distinguishing between strong binders like glucose and weaker binders like galactose and fructose. Radial distribution function analysis, supported by GIST, revealed the important role of water displacement in the binding process. The role of water in stabilizing complexes through H-bonding was also revealed, particularly in the case of glucose where there seems to be a complementary effect of water, providing around 4 H-bonds to the bound guest. And finally, hydrogen bonding between the host and the guest is the predominant driving force for complex formation. Glucose forms more hydrogen bonds with the receptor, an average of 5.5, than galactose which forms only 4 H-bonds. These findings not only reproduce experimental selectivity trends but also, through detailed analyses of hydrogen bonding, noncovalent interactions, and solvation effects, reveal the subtle mechanisms driving GluHUT's profound selectivity for glucose.

In the longer term, these studies may serve as a basis for predicting the behaviour of new receptor designs. The lack of reliable methods for assessing designs presents a major obstacle to progress in biomimetic carbohydrate recognition, and to work in related areas. Standard desktop modelling packages have proved of limited use, probably because they rely on continuum solvation and estimate energies rather than free energies. Methods based on MM MD could provide a way forward, greatly extending the scope of receptor design.

Conflicts of interest

There are no conflicts to declare.

Data availability

Computational methods are within the Supporting Information, along with additional references.

Acknowledgements

M.H. and A.J.M. thank the Engineering and Physical Sciences Research Council (EPSRC) for funding (grant number EP/W013738/1). R.E., A.J.M. and A.P.D. acknowledge a UKRI Frontier Research Grant selected by the ERC and funded by the Engineering and Physical Sciences Research Council (grant number EP/Y027779/1). This work was carried out using the computational and data storage facilities of the Advanced Computing Research Centre, University of Bristol – http://www.bristol.ac.uk/acrc/.

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